11 kV FROM HV SWITCHBOARD
F12-SW-012
2000-F12-SLD-EL-10001

F12-SR-002
415V PRIMARY CRUSHING SWITCHROOM

F12-TR-001
TRANSFORMER ROOM

F12-TR-001
11/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

4000/ 1A
2.5VA
5P20

F12-SW-001
PRIMARY CRUSHING 415V MOTOR CONTROL CENTRE

NOTE 1
1/700A
4000/ 5A
5VA
5P20

00F-01
4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
TEST
TEST
00F-08
ACB
AFD
4000A
240VAC SPRING
CHARGE SUPPLY
1F-01
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 4000A 65kA for 1sec

3F-01	3R-01	2R-09	3R-09	1R-05	1R-01	1F-13	1R-15	3F-08	3F-12	3F-20	2F-09
20A	160A	630A / 1000A	630A / 1000A	100A	20A	630A	400A	20A	20A	100A	32A
2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A	2A / 32A
70A	90A	100/1A	100/1A	90A	70A	400A	100/1A	70A	70A	70A	70A
EOL		EOL		EOL	EOL		EOL	EOL	EOL	EOL	EOL
M 3~	CP	RTD M 3~	RTD M 3~	t° M 3~	M 3~	t° M 3~	M 3~	M 3~	M 3~	M 3~	
1.5kW	52.48kW	355kW	355kW	37kW	0.75kW	132kW	1.1kW	1.1kW	37kW	8kW	
F12-BW-401 PRIMARY CRUSHER 2 DUST SEAL BLOWER 2000-F12-SCM-EL-10209	F12-CU-401 PRIMARY CRUSHER 2 CHILLER 2000-F12-SCM-EL-10211	F12-CV-401 PRIMARY CRUSHER 2 DISCHARGE CONVEYOR 2000-F12-SCM-EL-10401	F12-CV-402 STOCKPILE FEED CONVEYOR 2000-F12-SCM-EL-10402	F12-FN-403 PRIMARY CRUSHER 2 DUST COLLECTOR FAN 2000-F12-SCM-EL-10212	F12-VR-401 PRIMARY CRUSHER 2 DUST COLLECTOR ROTARY VALVE 2000-F12-SCM-EL-10407	F12-FA-401 PRIMARY CRUSHER 2 DISCHARGE APRON FEEDER 2000-F12-SCM-EL-10213	F12-FN-421 PRIMARY CRUSHER 2 DISCHARGE APRON FEEDER GEARBOX COOLER FAN 2000-F12-SCM-EL-10214	F12-PO-436 PRIMARY CRUSHER 2 DISCHARGE APRON FEEDER GEARBOX COOLER OIL PUMP 2000-F12-SCM-EL-10215	F12-FN-410 PRIMARY CRUSHER 2 BURIED VAULT VENTILATION FAN 2000-F12-SCM-EL-10217	F12-HT-407 to 410 PRIMARY CRUSHER 2 LUBRICATION RESERVIOR HEATERS 1 - 4 2000-F12-SCM-EL-10218	

FEEDER TO F12-SW-101
2000-F12-SLD-EL-10211
2000-F12-SCM-EL-10232

E-SHUT
F12SR001-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F12-SLD-EL-10202

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F12-SW-001 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F12-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F12-SLD-EL-10201_Rev2

Primary crushing F12-SW-001 415V MCC single line diagram sheet 1